AIMCR FORENSIC REPORT

AI-Powered Mobile Forensic Intelligence Platform for Autonomous Crime Reconstruction

Case Reference:	crime2
Case Status:	Active
Date Generated:	2026-08-20 19:52:20
Assigned Investigator:	Lead Investigator John
Evidence Count:	51 artifacts ingested
Forensic Complexity Index:	80/100 (Critical)

Executive Summary

This document compiles explainable investigative findings extracted from mobile device artifacts using the Autonomous Crime Reconstruction and Behavioral Profiling engines. Every finding detailed within this report is traceably backed by a corresponding SHA-256 hashed forensic artifact, maintaining full chain of custody. The contents represent investigative directions and indicators of probability rather than definitive judicial outcomes.

PART I: FORENSIC FACTS (DETERMINISTIC ANALYSIS)

The following sections compile deterministic logs, anomalies, anti-forensics alerts, and modular graphs directly extracted from device artifacts. All forensic findings are backed by SHA-256 integrity hashes.

1. Incident Reconstruction

No structured cybercrime sequences were reconstructed by the engine.

2. Behavioral Anomaly Log

The Isolation Forest engine analyzed the user's historical feature vectors (active hours, GPS coordinates, application categories, file extensions) and highlighted the following deviations from baseline behavior:

Timestamp	Type	Score	Anomaly Explanation / Context	Ref
2026-07-20 18:15:00	GPS	1.00	Flagged location anomaly at coordinates (34.0407, -118.2684). This location is far outside the user's standard historical geographic cluster.	7e716662
2026-08-05 08:25:00	GPS	0.96	Flagged location anomaly at coordinates (34.0522, -118.2437). This location is far outside the user's standard historical geographic cluster.	0124192d
2026-07-20 18:00:00	GPS	0.95	Flagged location anomaly at coordinates (34.0522, -118.2437). This location is far outside the user's standard historical geographic cluster.	f90181c3
2026-08-05 08:10:00	GPS	0.89	Flagged location anomaly at coordinates (40.7128, -74.006). This location is far outside the user's standard historical geographic cluster.	87d47f4c
2026-08-05 08:20:00	GPS	0.89	Flagged location anomaly at coordinates (40.713, -74.0058). This location is far outside the user's standard historical geographic cluster.	6df8bdb5
2026-07-10 11:00:00	Application	0.77	Application 'Telegram' was launched at an atypical hour.	f636b12a
2026-07-20 19:40:00	WiFi	0.63	Behavioral baseline deviation detected based on multi-variate event clustering.	c17626db
2026-08-20 19:38:28	SystemLogs	0.61	Behavioral baseline deviation detected based on multi-variate event clustering.	a9144476

2a. Anti-Forensics & Tampering Log

Heuristic scanners inspected the integrity of the device artifacts to find evidence of active concealment, extension spoofing, file system timestamp manipulation, or database tampering:

No active anti-forensics attempts or database anomalies were flagged.

3. Relationship Network & Priority Entities

The communication graph maps contacts, browser domains, and apps into connected modular communities. Nodes are assigned an Investigative Priority score using degree and betweenness centralities.

Priority Investigation Targets

Entity Identifier	Node Class	Investigative Priority Score	Action Needed
secure-auth-bank.net	Location	85.0/100	Immediate Forensics Review
GPS Location Logger	Application	20.67/100	Monitor & Validate
Coords: 34.052, -118.244	Location	6.0/100	Monitor & Validate
Coords: 34.041, -118.268	Location	6.0/100	Monitor & Validate
Coords: 40.713, -74.006	Location	6.0/100	Monitor & Validate
Chrome Browser	Application	6.0/100	Monitor & Validate

3a. Forensic Investigation Recommendations

AI/Deterministic Recommendations Summary: Based on the analysis of active threat intelligence matches and anti-forensics indicators, the investigator should prioritize disassembling the spoofed APK binary (disguised as a PNG) and carving the tampered SQLite database. Reconstructing network infrastructure and verifying modification timestamps via system logs is critical to confirm the attack vector and trace perpetrator activity.

Recommended Actions	Priority	Forensic Rationale / Next Steps
Perform WHOIS and passive DNS resolution query on 'secure-auth-bank.net'	High	The phishing infrastructure domain/URL 'secure-auth-bank.net' was matched. This will identify domain registrars, hosting IPs, and subdomains.
Check proxy/DNS resolution logs on client device for contacts with 'secure-auth-bank.net'	Medium	Determines exact duration and timeline of client communication with the malicious infrastructure.

4. Evidence Integrity Index (Chain of Custody)

Every artifact imported into the platform is indexed below with its original acquisition timestamp and a SHA-256 integrity hash to ensure transparency and auditability.

Evidence ID	Timestamp	Type	Source File	SHA-256 Hash Signature
ca8ec056...	2026-07-01 12:00:00	Application	installed_packages.json	75c56e7f92da9d69...
f636b12a...	2026-07-10 11:00:00	Application	installed_packages.json	75c56e7f92da9d69...
6f9e7fe5...	2026-07-15 18:22:00	Application	installed_packages.json	75c56e7f92da9d69...
f775994a...	2026-07-20 09:12:00	Call	calls.csv	55a9fcd44b1a867c...

cf4371bd...	2026-07-20 10:00:00	WiFi	wifi_connections.json	3ad7ec500be77f3f...
a8b1d17a...	2026-07-20 13:40:00	Call	calls.csv	55a9fcd44b1a867c...
e51231a4...	2026-07-20 14:30:00	SMS	sms_backup.json	189ffaa1719e1df2...
b1fcc289...	2026-07-20 14:32:00	SMS	sms_backup.json	189ffaa1719e1df2...
0d71c587...	2026-07-20 15:10:00	SMS	sms_backup.json	189ffaa1719e1df2...
7912b242...	2026-07-20 17:15:00	Call	calls.csv	55a9fcd44b1a867c...
f90181c3...	2026-07-20 18:00:00	GPS	gps_route.xml	11bdb4b4c71cfeba...
7e716662...	2026-07-20 18:15:00	GPS	gps_route.xml	11bdb4b4c71cfeba...
c17626db...	2026-07-20 19:40:00	WiFi	wifi_connections.json	3ad7ec500be77f3f...
e7b8d9fa...	2026-08-05 03:58:40	Contact	contacts.json	8e261bede328f224...
95d909c0...	2026-08-05 03:58:40	Contact	contacts.json	8e261bede328f224...
f6b9ecb8...	2026-08-05 03:58:40	Contact	contacts.json	8e261bede328f224...
785f6371...	2026-08-05 03:58:40	DeviceInformation	device_info.json	148a8f2cfc0c2115...
2ba16594...	2026-08-05 03:58:40	BrowserHistory	wifi_history.json	1b9a61588b329133...
50b6c269...	2026-08-05 03:58:40	BrowserHistory	wifi_history.json	1b9a61588b329133...
e6194c4d...	2026-08-05 07:30:00	Call	calllogs.json	1f418020913b7d51...
3a6d15e6...	2026-08-05 08:00:00	SMS	sms_messages.json	c635e3c698d77702...
f0359c95...	2026-08-05 08:05:00	SMS	sms_messages.json	c635e3c698d77702...
87d47f4c...	2026-08-05 08:10:00	GPS	gps_locations.json	e0b1cda71b7ed6f3...
bfd4b8ee...	2026-08-05 08:15:00	TimelineEvent	case_timeline.json	b9d3c4f82bbf775d...
b8868f2b...	2026-08-05 08:15:00	SMS	sms_messages.json	c635e3c698d77702...

** Showing 25 of 51 total evidence records. Full trace available in database.*

PART II: AI-ASSISTED INTERPRETATION (NON-DETERMINISTIC SUMMARY)

Important Notice: The content in this section is automatically generated by a local, on-premise Large Language Model using grounded context from parsed device evidence. It is intended for interpretation and intelligence-guided search assistance. This summary is non-deterministic and must be independently verified by a human forensic examiner. It does not replace original forensic facts.

[AI investigation assistance is currently unavailable. Falling back to local rules]

I retrieved the following relevant evidence records to answer your question:

- [2026-07-20 14:30:00] (SMS ID: e51231a4-8714-4e36-bb99-d4c8671042c7) -> SMS (incoming) from '+15550011' to 'Me': "Hey, meet at the library tonight"
- [2026-07-20 14:32:00] (SMS ID: b1fcc289-0887-4aaa-9825-2d82f03a911e) -> SMS (incoming) from 'Me' to '+15550011': "Understood, bringing the drive"
- [2026-07-20 15:10:00] (SMS ID: 0d71c587-088f-4a45-87c5-f92819cc368a) -> SMS (incoming) from '+15550022' to 'Me': "Did you access the server? Code is 9982"
- [2026-08-05 08:00:00] (SMS ID: 3a6d15e6-cd24-4607-b864-4aab62b2812a) -> SMS (incoming) from '+12125550110' to 'Me': "Hey honey, call me when you can!"
- [2026-08-05 08:05:00] (SMS ID: f0359c95-f862-42da-a758-69cdfca1e905) -> SMS (incoming) from '+12125550120' to 'Me': "Sent over the weekly brief report."
- [2026-08-05 08:15:00] (SMS ID: b8868f2b-ace2-46d0-b8cd-9a1634aa3bdf) -> SMS (incoming) from '+18005550199' to 'Me': "CRITICAL: Your bank account is suspended. Verify immediately at http://secure-auth-bank.net/login"
- [2026-08-05 08:21:00] (SMS ID: 14911dc7-e05d-46e7-9970-20edec6b26bd) -> SMS (incoming) from 'Bank Alerts' to 'Me': "Your OTP code for transfer of $5,000.00 is 882910. NEVER share this code with anyone."
- [2026-08-05 09:28:40] (File ID: a3a1d2e7-2722-4ae6-9b1b-e50e14775596) -> File modification: 'bank_alert_notice.pdf' in '' (15 bytes)
- [2026-08-05 09:28:40] (File ID: 773e372f-d664-4857-bf62-a71cd82732f2) -> File modification: 'phishing_domain_photo.jpg' in '' (111 bytes)
- [2026-08-05 09:28:40] (File ID: bae57882-6db6-486f-a293-857a50422f25) -> File modification: 'suspicious_screenshot.png' in '' (108 bytes)

If these logs do not directly answer your question, please refine your search queries or upload additional artifacts.

AI Interpretation Confidence: Low (Local Fallback)

Evidence Gaps / Limitations Identified by AI: Ollama service is currently unavailable or timed out., Analysis performed using deterministic keyword heuristics.

PART III: INVESTIGATOR OBSERVATIONS (MANUAL REMARKS)

This section records manual remarks, case hypothesis, and initial case notes entered by the assigned investigator.

Initial Case Notes / Objective:

No case description provided by the investigator.

Examiner Observations & Log Notes:

5. System Investigation Audit Trail

Immutable audit log records detailing investigator queries, report configurations, and analyses run inside this case file:

Timestamp	Action	Investigator details / Queries
2026-08-20 14:08:28	ACQUIRE_DEVICE_SUCCESS	Successfully acquired device: pixel_6_mock_001. Imported 21 records.
2026-08-20 14:08:24	ACQUIRE_DEVICE	Initiated forensic acquisition for device: pixel_6_mock_001 in case ID: 94b3b59a-aefc-4dcd-bc20-79709aabfaa6
2026-08-20 13:59:15	LOGIN	User investigator@aimcr.gov logged in successfully
2026-08-19 16:28:58	LOGIN	User investigator@aimcr.gov logged in successfully
2026-08-17 16:57:59	QUERY_AI_ASSISTANT	Queried AI Assistant: 'Summarize this investigation case.' on case ID '94b3b59a-aefc-4dcd-bc20-79709aabfaa6'
2026-08-17 16:57:39	LOGIN	User admin@aimcr.gov logged in successfully
2026-08-17 16:57:37	LOGIN	User investigator@aimcr.gov logged in successfully
2026-08-17 16:56:33	LOGIN	User investigator@aimcr.gov logged in successfully
2026-08-17 16:55:39	LOGIN	User investigator@aimcr.gov logged in successfully
2026-08-17 16:55:30	LOGIN	User investigator@aimcr.gov logged in successfully